Software Engineering Laboratory Project Proposal

AI Resume Analyzer and Search Hub

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1 Project Overview

Recruitment has become increasingly competitive due to the large number of applications submitted for every job opening. Recruiters often spend significant time manually reviewing resumes to identify suitable candidates. This process is repetitive, time-consuming, and may lead to qualified applicants being overlooked because of missing keywords or improper resume formatting. On the other hand, many students and job seekers are unaware of whether their resumes meet industry standards or match the skills required for specific job roles.

The proposed project, **AI Resume Analyzer and Search Hub**, is a web-based intelligent platform designed to simplify the recruitment and job search process. The system combines Artificial Intelligence (AI), Natural Language Processing (NLP), and information retrieval techniques to analyze resumes, compare them with job descriptions, and provide detailed feedback.

The AI Resume Analyzer evaluates uploaded resumes by extracting important information such as technical skills, educational qualifications, work experience, certifications, and projects. It compares these details with the selected job description and generates a compatibility score. The system also identifies missing skills, suggests improvements, highlights formatting issues, and recommends keywords that can improve the resume's chances of passing Applicant Tracking Systems (ATS).

The Search Hub acts as a centralized search platform where users can explore job opportunities based on their skills, qualifications, preferred location, and interests. Instead of visiting multiple job portals individually, users can efficiently search and organize available opportunities in one place.

The proposed system is intended for students, fresh graduates, experienced professionals, recruiters, and HR departments. The expected outcome is a smart platform that saves time, improves resume quality, assists users in finding suitable jobs, and supports recruiters by providing a faster and more consistent initial screening process.

2 Problem Statement

The recruitment process today involves handling hundreds or even thousands of resumes for a single job vacancy. Human recruiters often face difficulty in reviewing every resume

thoroughly within limited time. As a result, qualified candidates may be rejected because their resumes do not contain the expected keywords or fail to meet Applicant Tracking System (ATS) requirements.

Students and fresh graduates also face challenges while preparing professional resumes. Most candidates are unable to determine whether their resumes effectively represent their skills, projects, certifications, and achievements. They often submit resumes without knowing how well they match a specific job description.

Existing resume analysis platforms mainly focus on grammar correction or resume templates and do not provide detailed AI-driven skill analysis. Similarly, users are required to visit multiple job portals separately while searching for employment opportunities, making the overall process inefficient and time-consuming.

Therefore, there is a need for an intelligent system that combines resume analysis with a centralized job search platform. The proposed AI Resume Analyzer and Search Hub addresses these limitations by providing automated resume evaluation, ATS score prediction, keyword analysis, skill-gap identification, resume improvement suggestions, and a unified job search interface. This integrated solution enhances both the candidate experience and recruiter productivity.

3 Objectives

The primary objectives of the proposed project are:

1. Develop a web-based AI Resume Analyzer and Search Hub using modern web technologies.
2. Allow users to securely register, log in, and manage their personal profiles.
3. Enable users to upload resumes in PDF or DOCX format.
4. Extract important information from resumes using Natural Language Processing techniques.
5. Compare uploaded resumes with selected job descriptions.
6. Generate an AI-based resume compatibility score.

7. Identify missing technical and soft skills required for the selected job.
8. Suggest resume improvements, formatting corrections, and ATS optimization techniques.
9. Recommend important keywords that increase resume visibility during recruitment.
10. Provide a centralized job search system where users can search opportunities using multiple filters.
11. Help recruiters perform faster initial resume screening using AI-generated analysis.
12. Improve overall recruitment efficiency while reducing manual effort.

4 Scope of the Project

The project focuses on building an intelligent web application that assists both job seekers and recruiters during the recruitment process.

Features Included

- User Registration and Secure Login
- User Profile Management
- Resume Upload (PDF and DOCX)
- Resume Parsing using AI
- Job Description Upload
- Resume and Job Description Comparison
- ATS Compatibility Score Generation
- Skill Gap Analysis
- Keyword Matching
- Resume Quality Evaluation

- Personalized Resume Improvement Suggestions
- Search Hub for Job Opportunities
- Job Search using Skills, Role, Company, and Location
- Downloadable Resume Analysis Report
- Responsive Web Interface

Features Excluded

The following functionalities are outside the scope of the current version of the project:

- Online Interview Scheduling
- Video Interview Analysis
- Employee Background Verification
- Direct Integration with External Recruitment Portals
- Mobile Application
- Multi-language Resume Analysis
- AI Voice Interview Assistant

Target Users

The proposed system is designed for multiple categories of users, including:

- Students preparing for internships and placements
- Fresh Graduates seeking employment
- Experienced Professionals searching for better opportunities
- Recruiters performing resume screening
- Human Resource (HR) Professionals
- Placement Cells of Educational Institutions

Platform

The system will be developed as a responsive **Web Application**. Users will be able to access the platform using modern web browsers on desktops, laptops, and tablets without installing additional software.

The application will consist of a frontend for user interaction, a backend for business logic and AI processing, and a database for storing user information, resumes, job descriptions, and analysis reports. Artificial Intelligence and Natural Language Processing modules will perform resume parsing, keyword extraction, skill matching, and recommendation generation.